

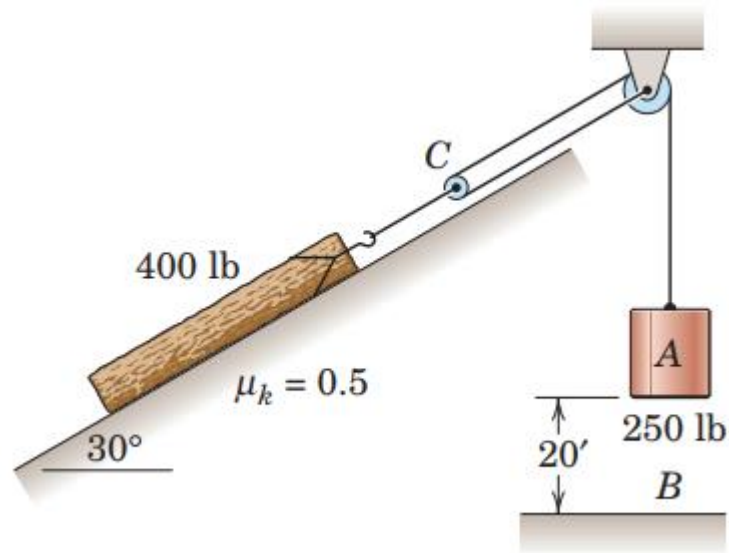
Kinetics of Particles

2nd Week

Dr. Atanu Banerjee

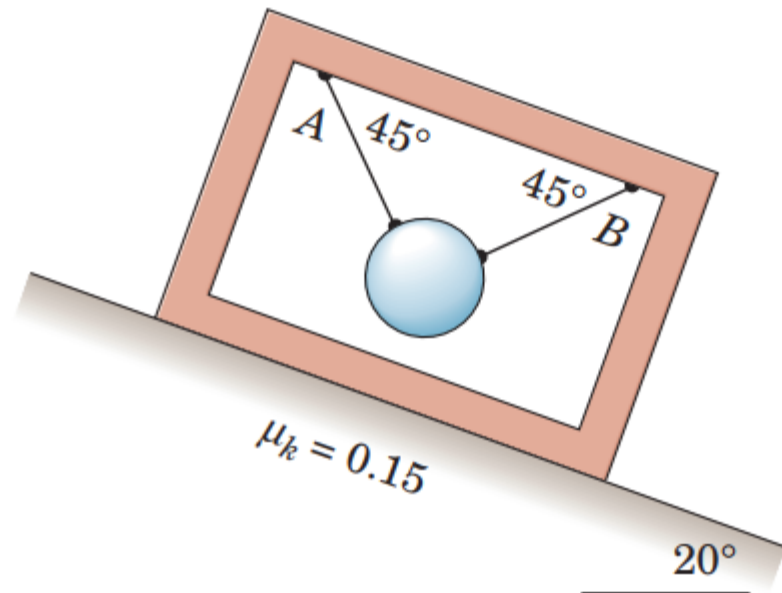
Example-1

The 250-lb concrete block *A* is released from rest in the position shown and pulls the 400-lb log up the 30° ramp. If the coefficient of kinetic friction between the log and the ramp is 0.5, determine the velocity of the block as it hits the ground at *B*.



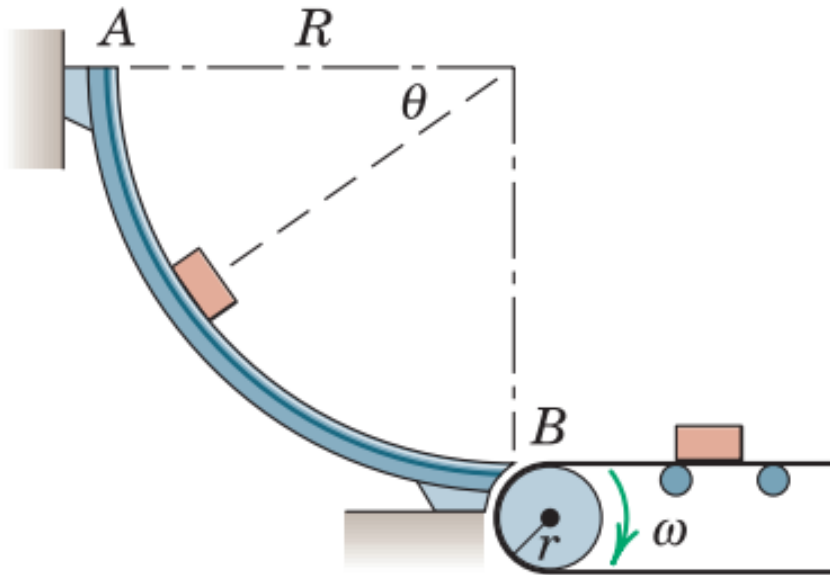
Example-2

The 10-kg steel sphere is suspended from the 15-kg frame which slides down the 20° incline. If the coefficient of kinetic friction between the frame and incline is 0.15, compute the tension in each of the supporting wires A and B .



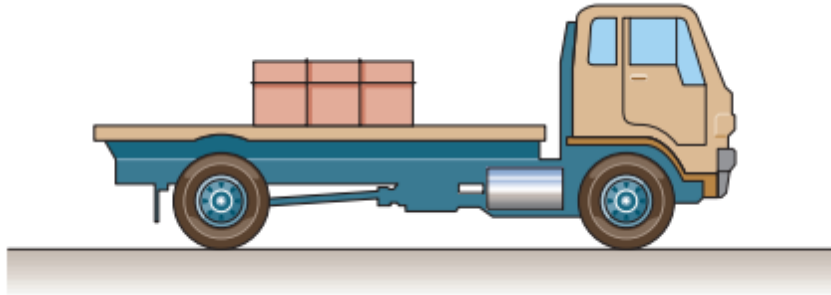
Example-3

Small objects are released from rest at A and slide down the smooth circular surface of radius R to a conveyor B . Determine the expression for the normal contact force N between the guide and each object in terms of θ and specify the correct angular velocity ω of the conveyor pulley of radius r to prevent any sliding on the belt as the objects transfer to the conveyor.



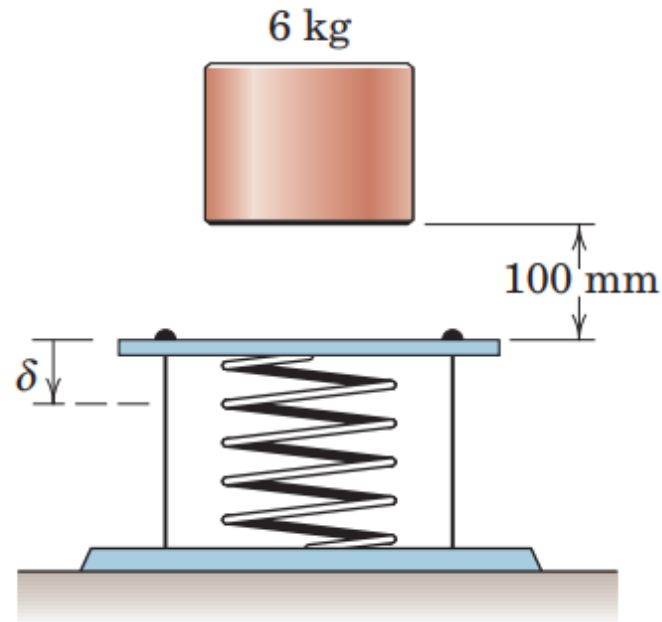
Example-4

The flatbed truck, which carries an 80-kg crate, starts from rest and attains a speed of 72 km/h in a distance of 75 m on a level road with constant acceleration. Calculate the work done by the friction force acting on the crate during this interval if the static and kinetic coefficients of friction between the crate and the truck bed are (a) 0.30 and 0.28, respectively, or (b) 0.25 and 0.20, respectively.



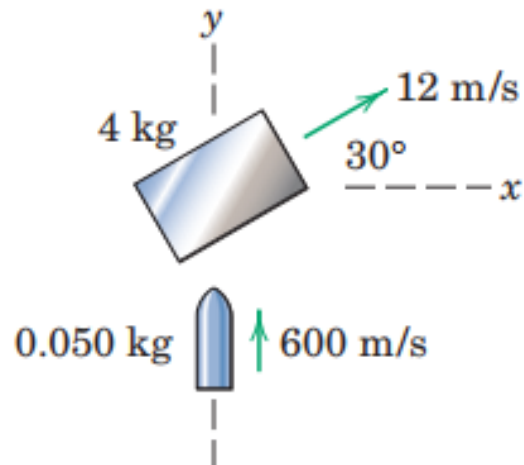
Example-5

The 6-kg cylinder is released from rest in the position shown and falls on the spring, which has been initially precompressed 50 mm by the light strap and restraining wires. If the stiffness of the spring is 4 kN/m, compute the additional deflection δ of the spring produced by the falling cylinder before it rebounds.



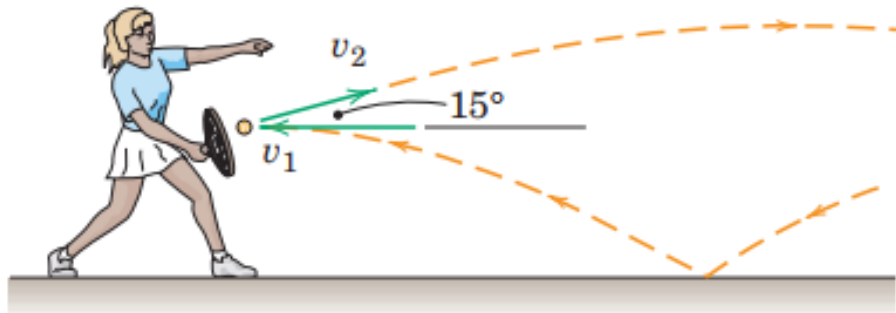
Example-6

The 50-g bullet traveling at 600 m/s strikes the 4-kg block centrally and is embedded within it. If the block slides on a smooth horizontal plane with a velocity of 12 m/s in the direction shown prior to impact, determine the velocity $\mathbf{v}_2$ of the block and embedded bullet immediately after impact.



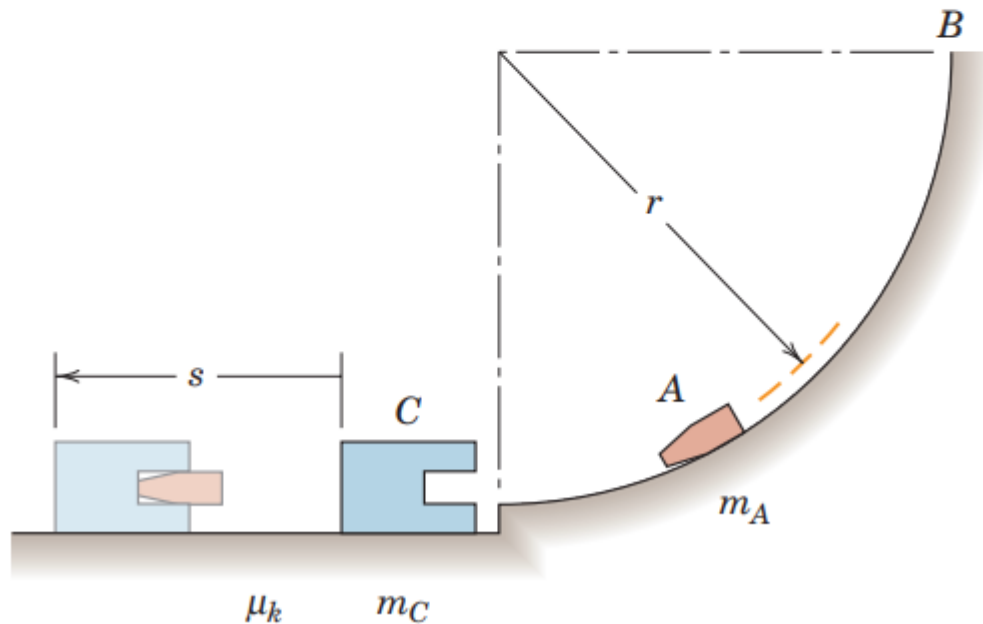
Example-7

A tennis player strikes the tennis ball with her racket when the ball is at the uppermost point of its trajectory as shown. The horizontal velocity of the ball just before impact with the racket is $v_1 = 50$ ft/sec, and just after impact its velocity is $v_2 = 70$ ft/sec directed at the 15° angle as shown. If the 2-oz ball is in contact with the racket for 0.02 sec, determine the magnitude of the average force $\mathbf{R}$ exerted by the racket on the ball. Also determine the angle β made by $\mathbf{R}$ with the horizontal.



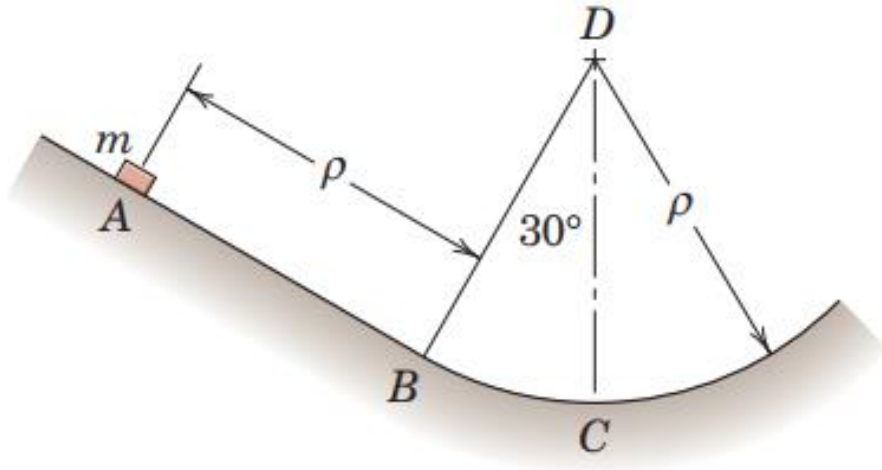
Example-8

The cylindrical plug A of mass m_A is released from rest at B and slides down the smooth circular guide. The plug strikes the block C and becomes embedded in it. Write the expression for the distance s which the block and plug slide before coming to rest. The coefficient of kinetic friction between the block and the horizontal surface is μ_k .



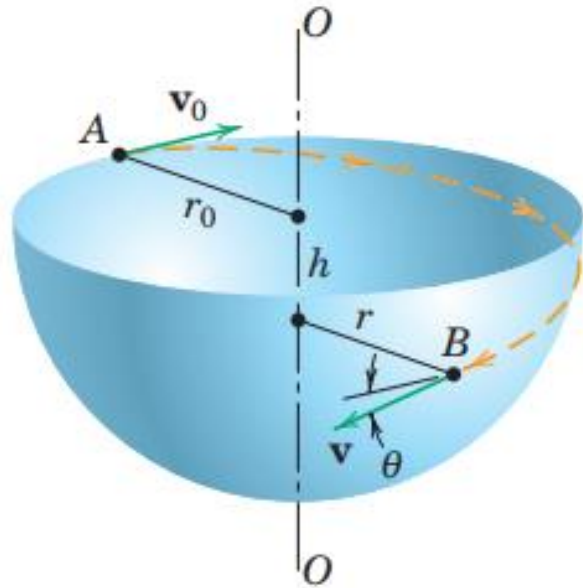
Example-9

A particle of mass m is released from rest in position A and then slides down the smooth vertical-plane track. Determine its angular momentum about both points A and D (a) as it passes position B and (b) as it passes position C .



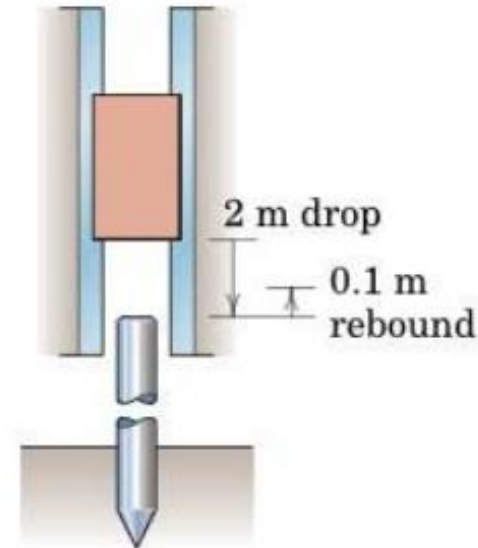
Example-10

A small mass particle is given an initial velocity $\mathbf{v}_0$ tangent to the horizontal rim of a smooth hemispherical bowl at a radius r_0 from the vertical centerline, as shown at point A . As the particle slides past point B , a distance h below A and a distance r from the vertical centerline, its velocity $\mathbf{v}$ makes an angle θ with the horizontal tangent to the bowl through B . Determine θ .



Example-11

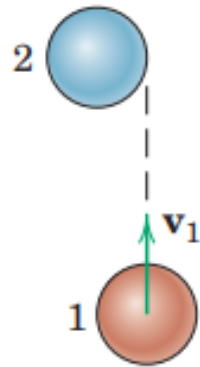
The ram of a pile driver has a mass of 800 kg and is released from rest 2 m above the top of the 2400-kg pile. If the ram rebounds to a height of 0.1 m after impact with the pile, calculate (a) the velocity v_p' of the pile immediately after impact, (b) the coefficient of restitution e , and (c) the percentage loss of energy due to the impact.



Example-12

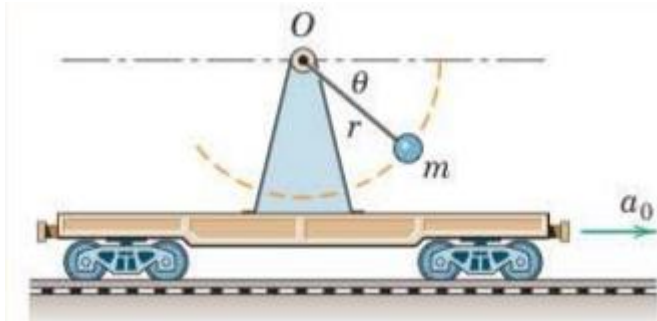
Oblique Central Impact Problem

Spherical particle 1 has a velocity $v_1 = 6$ m/s in the direction shown and collides with spherical particle 2 of equal mass and diameter and initially at rest. If the coefficient of restitution for these conditions is $e = 0.6$, determine the resulting motion of each particle following impact. Also calculate the percentage loss of energy due to the impact.



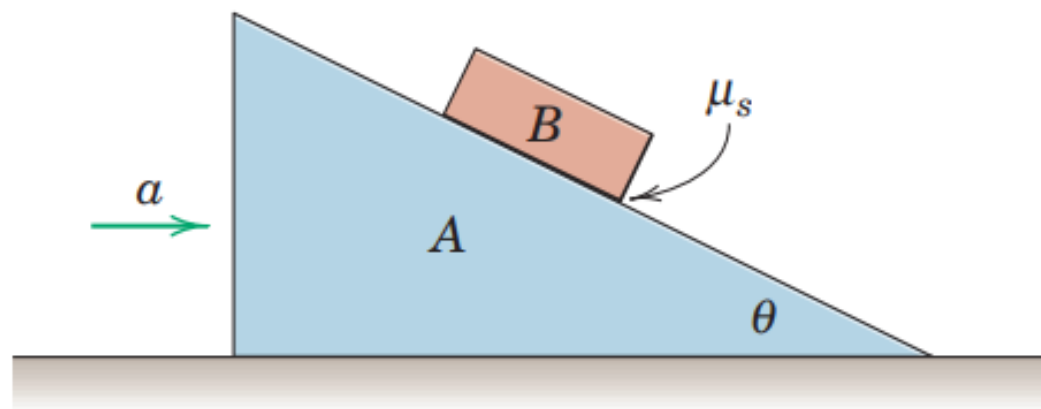
Example-13

A simple pendulum of mass m and length r is mounted on the flatcar, which has a constant horizontal acceleration a_0 as shown. If the pendulum is released from rest relative to the flatcar at the position $\theta = 0$, determine the expression for the tension T in the supporting light rod for any value of θ . Also find T for $\theta = \pi/2$ and $\theta = \pi$.



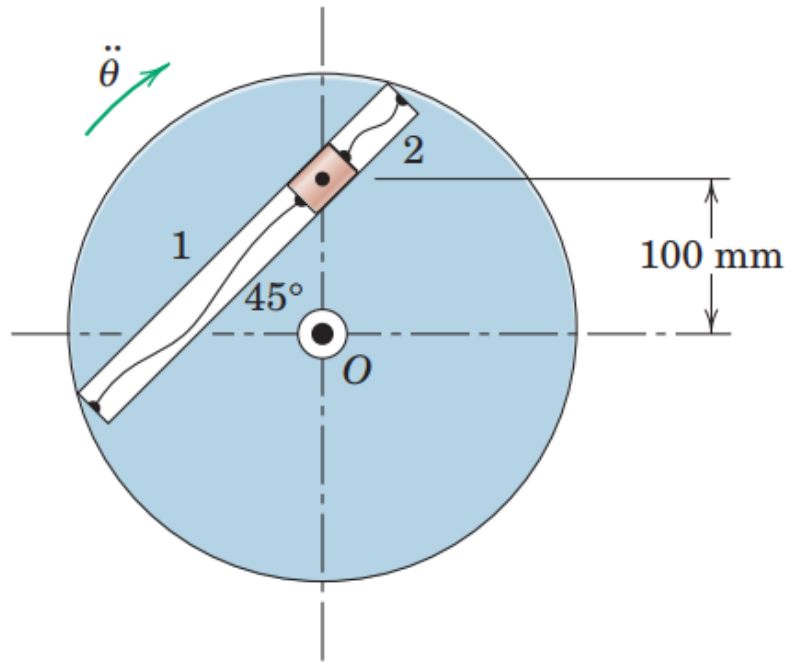
Example-14

The inclined block A is given a constant rightward acceleration a . Determine the range of values of θ for which block B will not slip relative to block A , regardless of how large the acceleration a is. The coefficient of static friction between the blocks is μ_s .



Example-15

The 2-kg slider fits loosely in the smooth slot of the disk, which rotates about a vertical axis through point O . The slider is free to move slightly along the slot before one of the wires becomes taut. If the disk starts from rest at time $t = 0$ and has a constant clockwise angular acceleration of 0.5 rad/s^2 , plot the tensions in wires 1 and 2 and the magnitude N of the force normal to the slot as functions of time t for the interval $0 \leq t \leq 5 \text{ s}$.



Example-16

The spring attached to the 10-kg mass is nonlinear, having the force–deflection relationship shown in the figure, and is unstretched when $x = 0$. If the mass is moved to the position $x = 100$ mm and released from rest, determine its velocity v when $x = 0$. Determine the corresponding velocity v' if the spring were linear according to $F = 4x$, where x is in meters and the force F is in kilonewtons.

